BT4012 Fraud Analytics

Academic Year 2026/2027 Semester 1 Lecturers: Dr. Yiliang ZHAO, Dr. Rui ZHAO

Project Guidelines and Grading Criteria

Updated as of 16th July 2026

Project Description

The course project is designed to provide hands-on experience in applying fraud analytics and machine learning techniques to address real-world fraud detection challenges. Working in groups of **3 to 4 students**, you are expected to design, develop, and evaluate a fraud detection pipeline using techniques taught in this module (and optionally beyond).

The project includes two deliverables:

- A **Group Project Presentation**
- A **Group Project Final Report**

Together, these components contribute **35% of your total course grade**.

Disclaimer:

Members of the same group will generally receive the same project marks. However, individual marks may be adjusted where there is credible evidence of substantial differences in contribution. All students will complete a confidential peer evaluation after the submission deadline. The teaching team may consider peer feedback, GitHub contribution history and other relevant evidence when determining whether an individual adjustment is appropriate. Students should raise serious contribution concerns as early as possible.

Important things to note:

Please name your files in the following format - report**XX**.pdf for Group Project Final Report, presentation**XX**.pptx for Group Project Presentation, and video**XX**.mp4 for the Presentation Video, where **XX** is your group number. For example, **Group 1**'s report should be named as

report01.pdf. Failure to follow the naming convention will result in a deduction equivalent to **2% of the relevant component's maximum mark**.

Project Component	**Due Date and Time**	**Submission Items**
Project Proposal [Optional]	20 Sept 2026@23:59	PDF Proposal
Group Project Presentation	20 Nov 2026@23:59	Presentation Deck and Video
Group Project Final Report	20 Nov 2026@23:59	PDF Report

Optional Project Proposal

When to submit: 20 Sept 2026, Sunday at 23:59
Who to submit: A representative of each group to submit
What to submit: 1 page PDF proposal
Where to submit: ***Canvas > Assignments > Project > Proposal***

You may submit a one-page project proposal by the stated deadline. Although optional, submission is highly encouraged if you would like early feedback on your project scope, dataset and methodology.

- Objective: To share your project idea so we can help assess its feasibility, data suitability, and technical scope.

- Format: 1 page outlining the problem statement, dataset(s), and your planned methodology (preliminary is fine).

This is not graded but intended to help you stay on track and avoid scope misalignment later in the semester.

Group Project Final Report

When to submit: 20 Nov 2026, Friday at 23:59
Who to submit: A representative of each group to submit
What to submit: PDF Report (Please include URL to your GitHub Repo in the report)
Where to submit: ***Canvas > Assignments > Project > Final Report***

Guidelines

You are to submit an **8–12-page PDF report** describing your completed project. The page limit excludes appendices and references.
Your GitHub repository must be accessible to the teaching team by the submission deadline and should include a README with instructions to reproduce your results.

Recommended Report Structure

1. **Problem Description**
    - Define the fraud problem your project addresses
    - Explain its real-world importance and inherent challenges

2. **Data Overview**
    - Source(s) of dataset(s)
    - Dataset characteristics (e.g., number of records, types of variables)
    - Tabular summary of variables

3. **Methodology**
    - Data pre-processing (e.g., cleaning, transformation, feature engineering)
    - Model selection and justification (e.g., supervised, unsupervised)
    - Analytical insights derived

4. **Experiments and Results**
    - Model performance evaluation (metrics, visualizations)
    - Integration into real-world business processes
    - Additional insights from experiments

5. **Conclusion**
    - Summary of findings
    - Limitations and future directions

Grading Criteria

The Group Project Final Report weighs **20%** of your overall final grade. Your report will be evaluated based on:

Criterion	**Marks**
Problem formulation and practical importance	2
Data understanding and preprocessing	3
Method selection and technical justification	4
Experimental design and evaluation	4
Results, insights and real-world applicability	3
Limitations and reproducibility	2
Clarity, organisation and conciseness	2
Total	**20**

Group Project Presentation

Who to present: Every member must participate meaningfully in the presentation, with speaking time distributed reasonably across the group.
What to present: ***a recorded presentation of no more than 15 minutes****, containing highlights of your analysis*
When to submit: 20 November 2026, Friday at 23:59
What to submit: Presentation Slides and the Presentation Video
Where to submit: ***Canvas > Assignments > Project > Presentation***

Guidelines

1. Prepare and record a group presentation of no more than 15 minutes. The presentation should highlight the most important elements of your analysis and may follow the general structure of the Final Report.
2. You must include the title and group members at the beginning of the video, either as the first page of your slides or as an overlay text. Make sure that you leave the title for long enough of a duration to be read (up to 8 seconds).
3. You may record your presentation by capturing your computer screen along with audio. Recommended screen recording tools include:
    a. For Windows users: OBS Studio, Camtasia
    b. For macOS users: OBS Studio, or the built-in QuickTime Player

    You can also record your presentation using Zoom, Microsoft Teams, or other online meeting platforms by starting a meeting and using the built-in recording function.
4. Make sure your final video has clear visuals and audio, and test that it plays correctly before submission.

Grading Criteria

The presentation will be graded out of 15 marks based on the following criteria. Submission of complete slides and a playable video is a basic submission requirement and does not automatically guarantee a minimum mark.

Criterion	**Marks**
Background and problem statement	2

Models and technical justification	3
Experiments, results and interpretation	3
Coherence and flow	2
Slide quality	2
Professionalism and delivery	2
Time limit and balanced contribution	1
Total	**15**

Finally, any use of generative AI tools, external code or public notebooks must be appropriately acknowledged, and students must be able to explain and justify all submitted work.

All the best for your project!

- End -